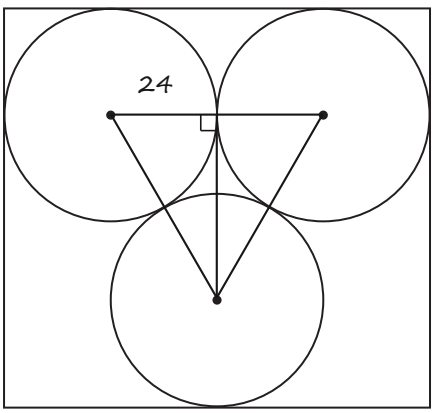
21 The diagram shows 3 identical circles inside a rectangle.
Each circle touches the other two circles and the sides of the rectangle, as shown in the diagram.



The radius of each circle is 24 mm.

Work out the area of the rectangle.
Give your answer correct to 3 significant figures.

Join the circle centres: each joining length is 48 mm.

Rectangle length = 4 × 24 = 96 mm

1 mark
Find the length

Let h be the perpendicular height of the centre triangle.

$h^2 = 48^2 - 24^2 = 1728$

$h = \sqrt{1728} = 24\sqrt{3} = 41.5692...$ mm

1 mark
Find the perpendicular height between centres

Rectangle height = 24 + h + 24 = 89.5692... mm

1 mark
Add a radius above and below the centre triangle

Area = $96 \times (48 + 24\sqrt{3}) = 8598.645...$ mm^2

To 3 significant figures: 8600 mm^2

1 mark
Correct area with appropriate rounding

8600 ..... mm^2

(Total for Question 21 is 4 marks)